

GUNA, India

WMO: 425590

Lat: 24.650N

Lon: 77.317E

Elev: 478

StdP: 95.71

Time Zone: 5.50 (IND)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.9	9.2	-0.2	3.9	20.8	1.1	4.3	19.1	3.2	17.1	2.5	18.2	0.2	50	0.309

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	13.6	43.0	22.9	41.8	22.5	40.6	22.2	27.3	32.7	26.8	31.7	26.4	31.1	1.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	22.9	29.4	25.7	22.2	28.8	25.3	21.7	28.2	90.4	32.8	88.1	31.7	86.1	31.1	33.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
3.0	2.4	2.2	DB	5.0	45.1	1.7	1.1	3.8	45.9	2.8	46.5	1.8	47.1	0.5	47.9	(4)
			WB	3.8	28.9	1.1	1.6	3.0	30.1	2.4	31.0	1.7	31.9	0.9	33.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	26.0	17.0	20.3	25.9	31.1	34.5	32.4	28.1	26.8	27.5	26.6	22.7	18.7	(6)
(7)		DBStd	5.72	2.52	2.95	3.04	2.47	1.97	3.28	2.27	1.45	1.62	1.87	2.01	2.42	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	98	58	13	0	0	0	0	0	0	0	0	1	26	(9)
(10)		CDD10.0	5830	217	288	492	632	758	672	560	520	526	516	381	268	(10)
(11)		CDD18.3	2887	17	68	233	382	500	422	302	262	276	257	132	36	(11)
(12)		CDH23.3	36129	191	664	2702	5612	8286	6353	3204	2279	2733	2601	1161	343	(12)
(13)		CDH26.7	20041	30	216	1479	3609	5856	4109	1261	618	1084	1296	418	63	(13)
(14)	Wind	WSAvg	0.6	0.5	0.6	0.6	0.6	0.8	0.9	0.7	0.6	0.5	0.3	0.3	0.4	(14)
(15)	Precipitation	PrecAvg	1077	10	10	13	3	10	147	377	327	147	15	8	12	(15)
(16)		PrecMax	1955	29	55	74	10	48	809	646	670	626	68	98	164	(16)
(17)		PrecMin	505	3	0	2	0	0	22	53	31	11	0	0	0	(17)
(18)		PrecStd	288	8	14	18	3	13	166	141	133	134	20	23	33	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.1	33.1	39.2	42.5	44.5	44.6	38.7	34.3	36.1	36.5	33.0	30.1	(19)
(20)		0.4%	MCWB	17.1	18.0	19.6	21.0	23.4	23.1	25.2	25.8	23.2	21.4	19.0	18.0	(20)
(21)		2%	DB	27.4	31.2	37.0	41.2	43.3	42.7	35.9	32.1	34.3	35.0	31.6	28.5	(21)
(22)		2%	MCWB	16.4	17.7	19.0	20.7	22.9	23.8	25.4	26.1	24.0	21.3	19.0	17.2	(22)
(23)		5%	DB	25.6	29.4	35.5	40.1	42.4	41.0	33.9	31.0	33.1	33.9	30.4	26.9	(23)
(24)		5%	MCWB	15.4	17.0	18.6	20.4	22.6	23.9	25.5	25.9	24.3	21.0	18.3	16.3	(24)
(25)		10%	DB	23.7	27.6	33.9	38.8	41.3	39.1	32.3	29.8	31.9	32.8	29.2	25.2	(25)
(26)		10%	MCWB	14.5	16.3	18.1	20.1	22.4	24.3	25.6	25.7	24.6	20.7	17.9	15.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.6	20.0	21.7	23.8	26.9	28.4	27.7	27.4	27.5	25.6	22.8	20.2	(27)
(28)		0.4%	MCDWB	26.9	28.1	34.6	39.5	38.9	36.5	32.8	30.7	31.1	30.8	27.7	27.8	(28)
(29)		2%	WB	17.3	18.7	20.4	22.2	25.7	27.3	27.2	26.8	26.6	24.5	21.3	18.2	(29)
(30)		2%	MCDWB	24.9	27.9	33.6	38.0	38.8	35.0	31.5	30.1	30.4	29.4	26.9	26.0	(30)
(31)		5%	WB	16.3	17.9	19.6	21.3	24.7	26.6	26.7	26.4	26.0	23.7	20.0	17.2	(31)
(32)		5%	MCDWB	23.3	27.2	32.4	37.1	38.1	34.3	30.6	29.5	30.0	28.7	27.3	25.1	(32)
(33)		10%	WB	15.4	17.0	18.8	20.6	23.9	26.0	26.3	26.0	25.5	22.8	19.1	16.2	(33)
(34)		10%	MCDWB	21.8	26.2	31.4	36.2	37.5	33.7	29.9	28.9	29.3	28.5	27.0	24.0	(34)
(35)	Mean Daily Temperature Range	MDBR	13.1	13.6	14.8	15.1	13.6	10.7	6.2	5.4	8.0	12.6	13.8	13.8	(35)	
(36)		5% DB	MCDBR	15.2	15.4	16.3	16.1	14.4	12.8	9.4	7.6	10.4	13.8	14.7	15.3	(36)
(37)		5% DB	MCWBR	6.8	6.2	5.5	5.1	4.7	3.9	2.9	2.9	3.2	4.4	5.7	6.5	(37)
(38)		5% WB	MCDWB	13.0	13.6	14.5	14.7	13.2	10.4	7.0	6.2	7.4	10.1	12.0	13.5	(38)
(39)		5% WB	MCWBR	6.4	6.2	5.7	5.4	5.2	3.8	3.1	2.9	3.2	4.0	5.3	6.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.439	0.438	0.469	0.531	0.589	0.660	0.646	0.583	0.521	0.519	0.519	0.475	(40)	
(41)		taud	2.031	2.078	2.013	1.895	1.812	1.719	1.767	1.928	2.054	1.944	1.875	1.944	(41)	
(42)		Ebn at Noon	800	838	832	788	739	684	693	738	776	750	713	740	(42)	
(43)		Edh at Noon	153	156	175	201	218	237	226	192	166	176	178	161	(43)	
(44)	All-Sky Solar Radiation	RadAvg	4.14	5.10	6.23	7.01	7.13	5.89	4.18	4.10	5.06	5.25	4.40	4.02	(44)	
(45)		RadStd	0.33	0.26	0.24	0.25	0.18	0.54	0.58	0.45	0.53	0.31	0.29	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	+2.05	N/A	+0.30	+0.43	N/A	N/A	N/A	N/A

Nomenclature: See separate page